SOFTWARE ENGINEERING ACADEMIC PROJECT — MID-SEMESTER EVALUATION

Department of Computer Science & Engineering

Project Meeting Logs & Team Agendas

Project: AutoTestHub (Web Automation & Testing Platform)

SPRINT STANDUPS • ARCHITECTURE REVIEWS • ACTION ITEM TRACKING

MEETING LOG RECORDS

Total Recorded Sprints: Sprints 1 to 3

Formal Meetings Conducted: 6 Sessions

Average Attendance: 100% (All 4 Members Present)

Meeting Modality: Hybrid (Lab Sessions & Online Sync)

PROJECT TEAM ATTENDEES

Shristi Neekhra (Meeting Facilitator)

Shifa Siddiqui (Backend Architect)

Shruti Gupta (Selenium Engine Architect)

Shriya Agarwal (Frontend Specialist)

Supervisor: [Supervisor Name]

Weekly Meeting Logs: Sprint 1 (Weeks 1 – 2)

Meeting #1: Project Inception & Problem Scoping

Date & Time	2026-09-08 • 15:30 – 17:00	**Location**	CS Department Project Lab / Hybrid
Attendees	Shristi Neekhra, Shifa Siddiqui, Shruti Gupta, Shriya Agarwal (100% Attendance)		
Meeting Objective	Select project domain, define problem statement, and allocate initial research responsibilities.		
Key Discussions	Discussed common industry bottlenecks in automated web regression testing. Identified lack of clean orchestration tools that bridge web UI, REST APIs, and headless browser workers without heavy enterprise bloat. Agreed to develop **AutoTestHub**.		
Key Decisions	1. Scope project as a multi-tier automated web testing platform using Selenium. 2. Divide architecture into 4 distinct modular areas: Frontend, Backend API, Selenium Engine, and Execution Bridge.		
Action Items Assigned	• Shristi: Draft initial WBS and repository structure. • Shifa: Research MongoDB schema designs for test cases and runs. • Shruti: Investigate Selenium 4 Selenium Manager and headless Chrome options. • Shriya Agarwal: Create wireframe concepts for the Control Desk dashboard.		

Meeting #2: Architectural Boundary & Interface Definition

Date & Time	2026-09-15 • 16:00 – 17:30	**Location**	Project Lab Room 302
Attendees	Shristi Neekhra, Shifa Siddiqui, Shruti Gupta, Shriya Agarwal		
Meeting Objective	Finalize technology stack and define the inter-process JSON communication contract.		
Key Discussions	Evaluated whether to communicate with the Python runner via HTTP microservice or standard child_process streams (stdin/stdout). Selected stdin/stdout IPC for zero network overhead, deterministic process lifecycle, and simple local execution.		
Key Decisions	1. Formalized JSON schema contract for test case definitions and output results. 2. Decided to enforce Mongoose schema validation on backend models.		
Action Items Assigned	• Shruti: Begin implementation of Python runner CLI and locator resolvers. • Shifa: Initialize Node.js Express server and setup MongoDB connection.		

Weekly Meeting Logs: Sprint 2 (Weeks 3 – 4)

Meeting #3: Backend Models & Action Handlers Review

Date & Time	2026-09-22 • 14:00 – 15:45	**Location**	Online Video Conference
Attendees	Shristi Neekhra, Shifa Siddiqui, Shruti Gupta, Shriya Agarwal		
Meeting Objective	Review completed MongoDB schemas and first iteration of Python action handlers.		
Key Discussions	Reviewed `TestCase` and `TestResult` Mongoose schemas. Shruti demonstrated `OPEN_URL` and `TYPE` action handlers running in Chrome with explicit waits.		
Key Decisions	1. Add automatic screenshot capture for both passing and failing runs. 2. Ensure all URLs strictly enforce HTTP/HTTPS schemes to prevent arbitrary file reading.		
Action Items Assigned	• Shifa: Implement `POST /api/tests` and `GET /api/tests` routes. • Shruti: Implement `ASSERT_TITLE` validator and screenshot capture utility. • Shriya Agarwal: Implement HTML/CSS responsive layout for Control Desk.		

Meeting #4: Integration Testing & UI Simulation Sync

Date & Time	2026-09-29 • 15:30 – 17:15	**Location**	Project Lab Room 302
Attendees	Shristi Neekhra, Shifa Siddiqui, Shruti Gupta, Shriya Agarwal		
Meeting Objective	Review interactive Control Desk prototype and in-memory test harness.		
Key Discussions	Shriya Agarwal demonstrated the interactive Control Desk with real-time log streaming and simulation mode. Shruti presented the 8 in-memory unit tests in `test_validation.py`.		
Key Decisions	1. Add a controlled local HTTP server integration test harness (`test_integration.py`) to test browser interactions without external network dependencies. 2. Begin work on the Node.js execution service spawner.		
Action Items Assigned	• Shristi: Begin writing `executionService.js` and standalone pipeline test. • Shruti: Finalize integration test harness with local HTTP server.		

Weekly Meeting Logs: Sprint 3 (Weeks 5 – 6)

Meeting #5: Execution Service & Pipeline Integration

Date & Time	2026-10-06 • 16:00 – 18:00	**Location**	CS Lab / Online Sync
Attendees	Shristi Neekhra, Shifa Siddiqui, Shruti Gupta, Shriya Agarwal		
Meeting Objective	Integrate Node.js execution service with Python worker and verify end-to-end flow.		
Key Discussions	Tested end-to-end execution where Express receives `POST /api/tests/:id/run`, fetches test case from MongoDB, spawns Python runner, and saves result. Debugged an issue where extraneous Chrome output caused JSON parsing errors in Node.js.		
Key Decisions	Applied a robust regex matcher (`/\{[\s\S]*\}/`) in Node.js to extract the clean JSON payload, resolving all parsing flakiness.		
Action Items Assigned	• Shristi: Verify standalone execution script and commit integration pipeline. • Shifa: Validate MongoDB connection error handling. • Shriya Agarwal: Add theme toggle and copy log buttons to Control Desk.		

Meeting #6: Mid-Semester Evaluation Package Assembly

Date & Time	2026-10-13 • 14:30 – 16:45	**Location**	CS Department Project Lab
Attendees	Shristi Neekhra, Shifa Siddiqui, Shruti Gupta, Shriya Agarwal		
Meeting Objective	Audit mid-semester deliverables against evaluation rubric, finalize documentation, and record demo video.		
Key Discussions	Audited project progress against guidelines: 50% completion achieved (surpassing 40-60% target). Verified 16/16 tests passing, screenshots generated, and git history updated. Rehearsed demo video script.		
Key Decisions	1. Package submission strictly adhering to the 7-folder recommended ZIP structure. 2. Include formal AI disclosure statement and signed contribution declaration in final report.		
Action Items Assigned	• All Members: Review individual contributions and confirm signed declaration. • Shristi: Generate final PDF artifacts and build submission ZIP.		